COMPUTER ENGINEERING

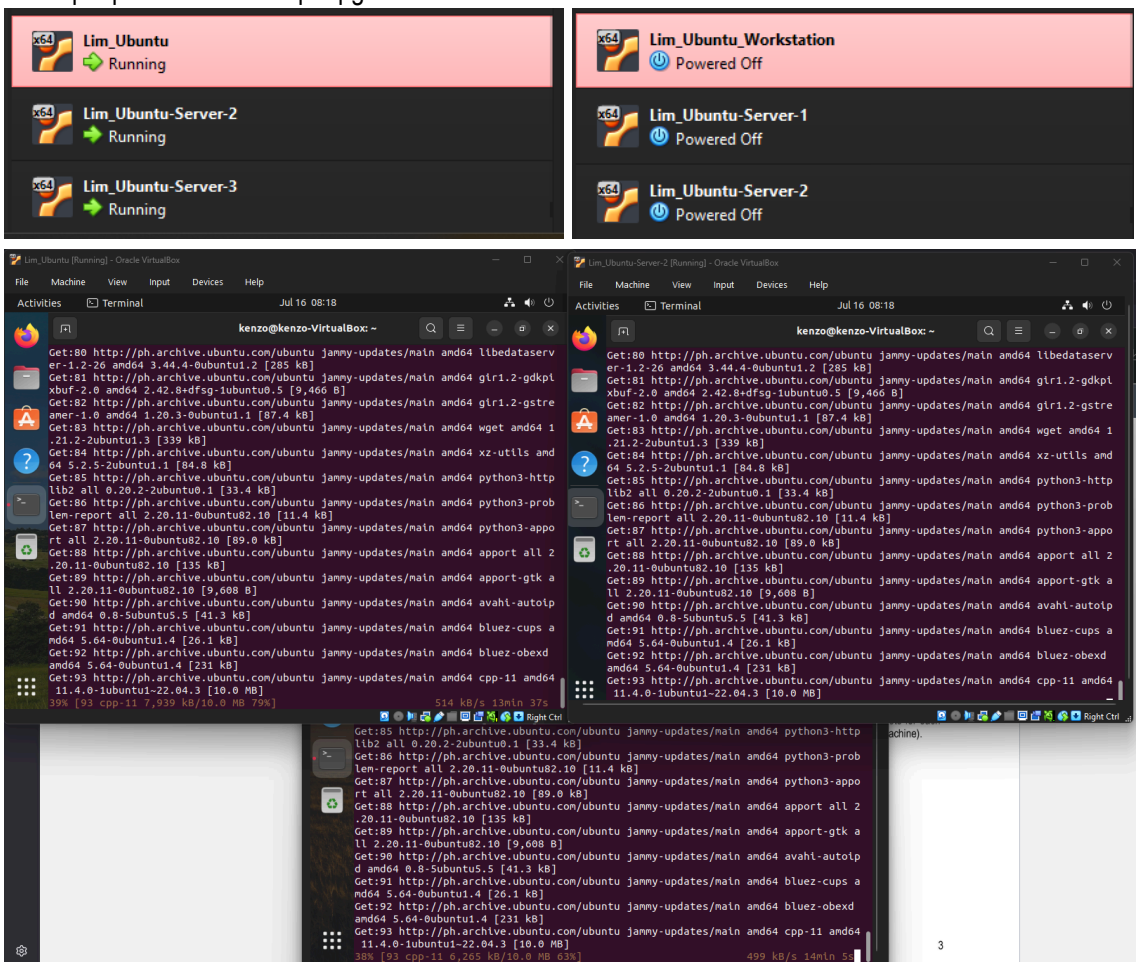
QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	7/16/2026	**DATE SUBMITTED:**	7/16/2026
NAME:	Kenzo N. Lim	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 1 Configure Networks using Virtual Machines

1. PROCEDURE OUTPUT

What procedures did you perform?
I first downloaded the Server 1 VM and cloned it for the 2nd and 3rd VM, after that i update each of the VMs using sudo apt update and sudo apt upgrade

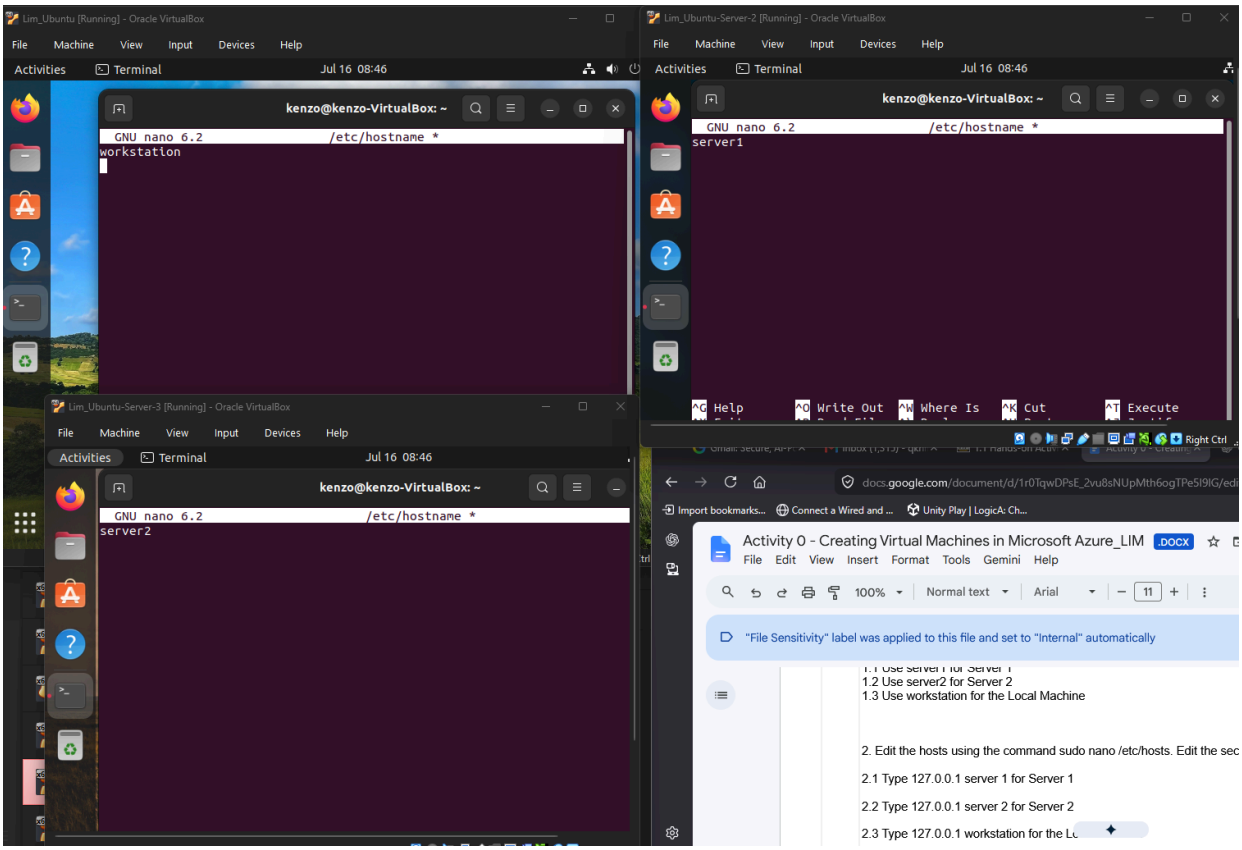


(REFER TO THIS IMAGE)

Task 1: Changing Hostname
Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano

command, press Control
+ O to write out (save the file). Press Enter when asked for the name of the file. Press Control
+ X to end.

1. Change the hostname using the command sudo nano /etc/hostname
1.1 Use server1 for Server 1
1.2 Use server2 for Server 2
1.3 Use workstation for the Local Machine

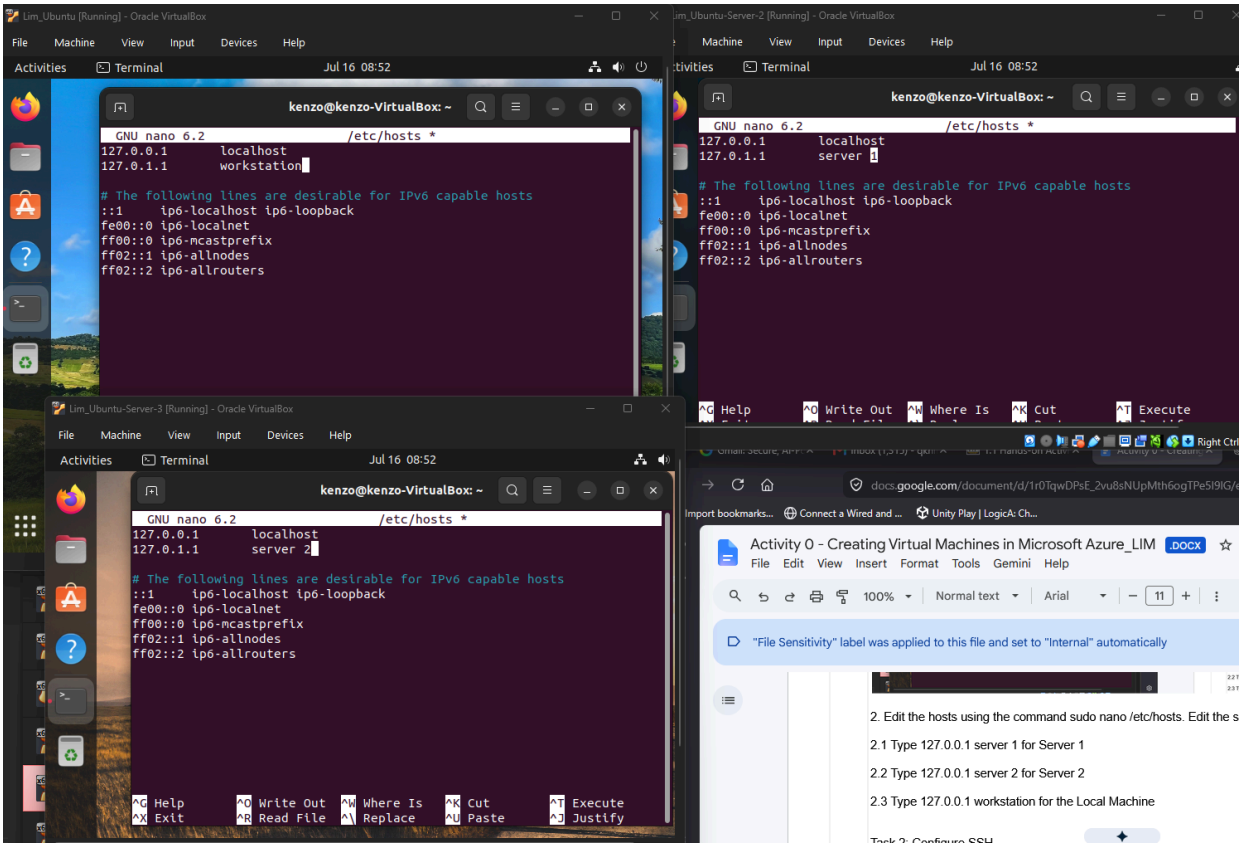


2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.

2.1 Type 127.0.0.1 server 1 for Server 1

2.2 Type 127.0.0.1 server 2 for Server 2

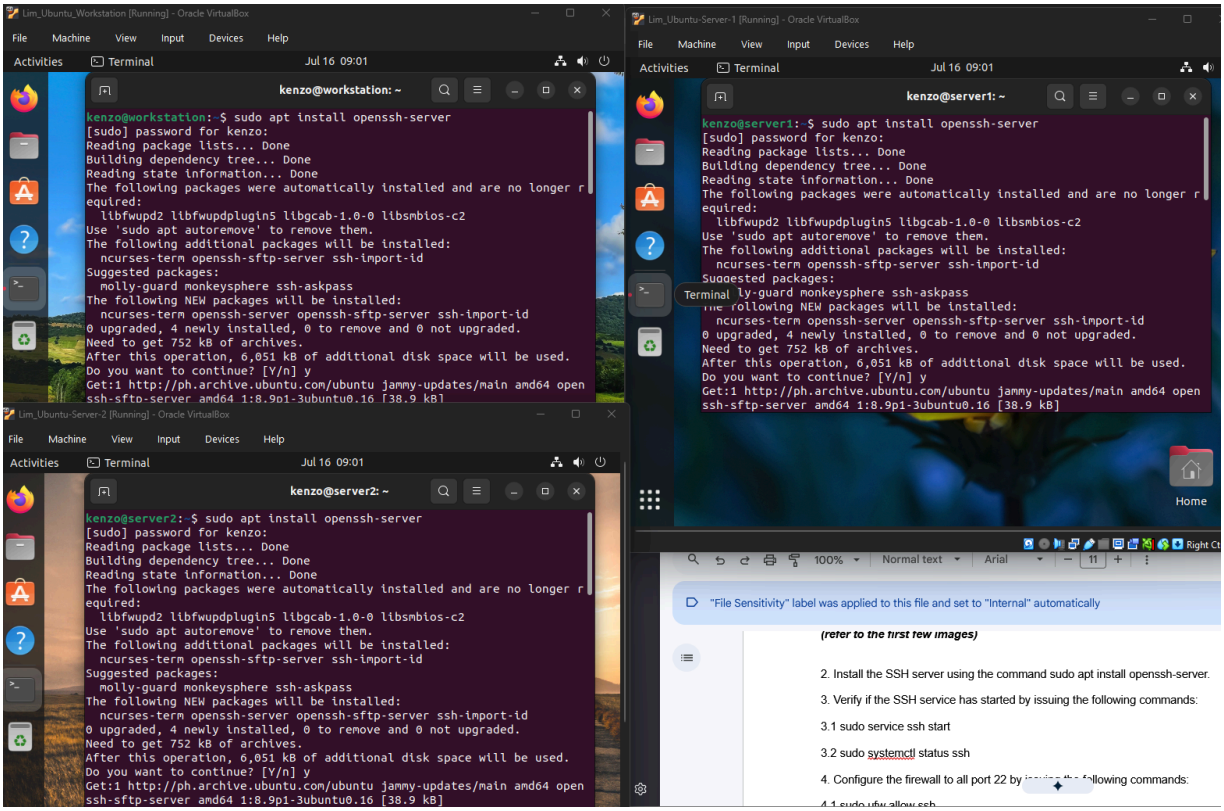
2.3 Type 127.0.0.1 workstation for the Local Machine



Task 2: Configure SSH
Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the commands sudo apt update and sudo apt upgrade, respectively.
(refer to the first few images)

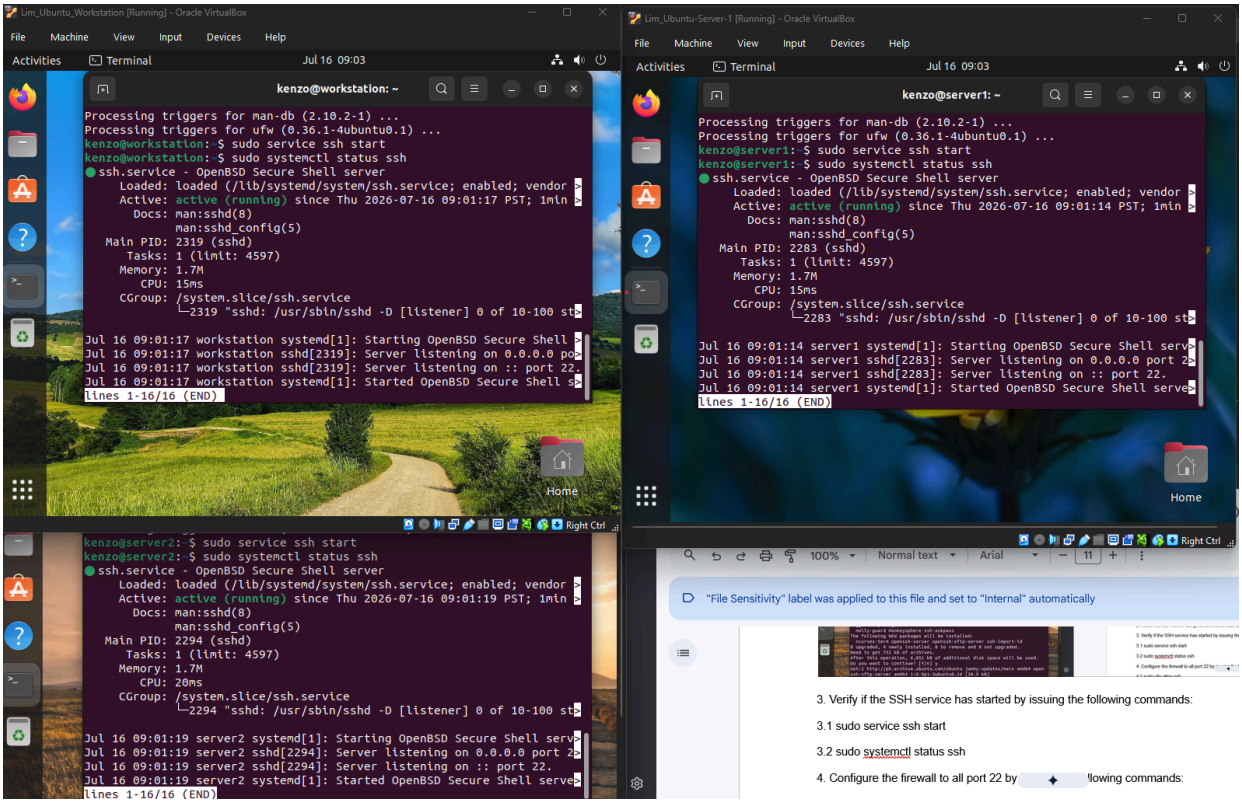
2. Install the SSH server using the command sudo apt install openssh-server.



3. Verify if the SSH service has started by issuing the following commands:

3.1 sudo service ssh start

3.2 sudo systemctl status ssh

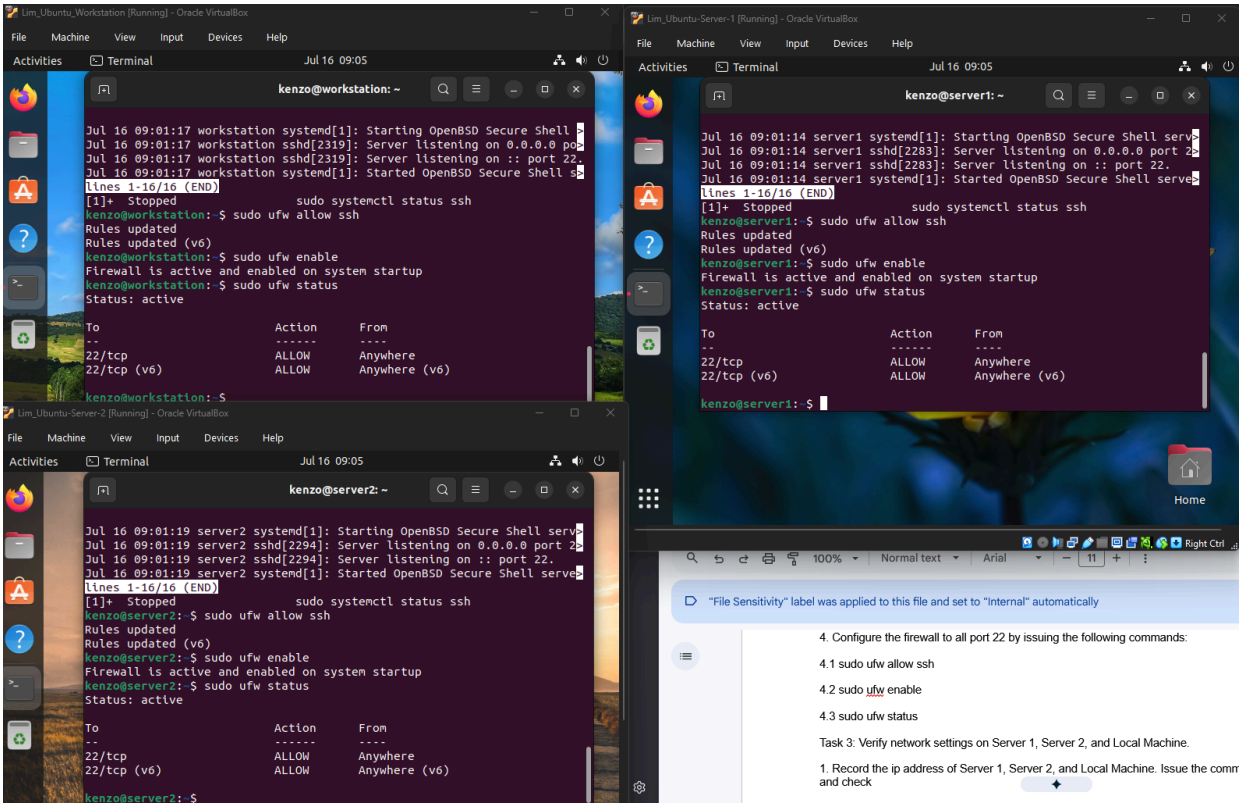


4. Configure the firewall to all port 22 by issuing the following commands:

4.1 sudo ufw allow ssh

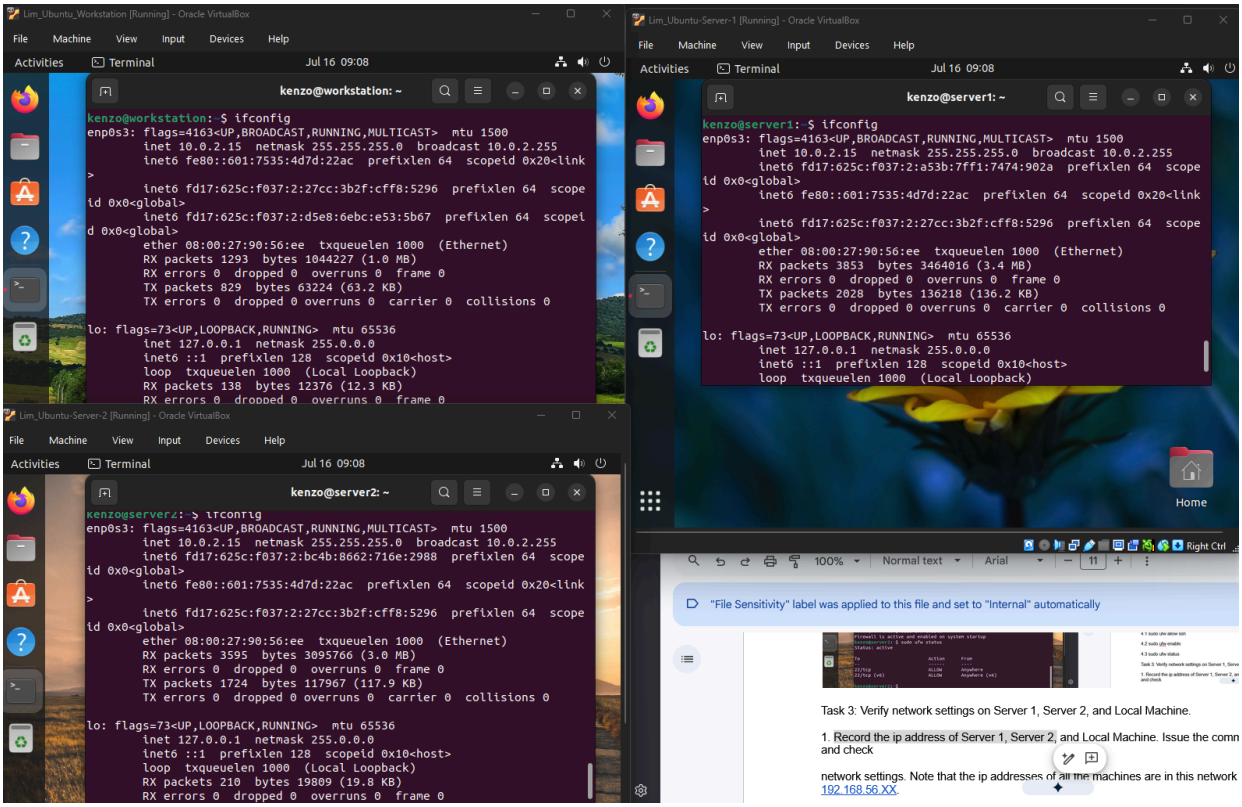
4.2 sudo ufw enable

4.3 sudo ufw status

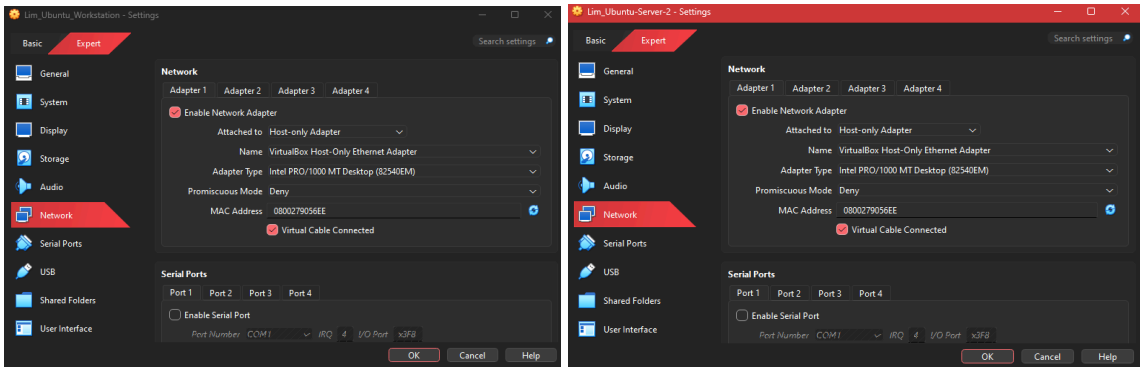


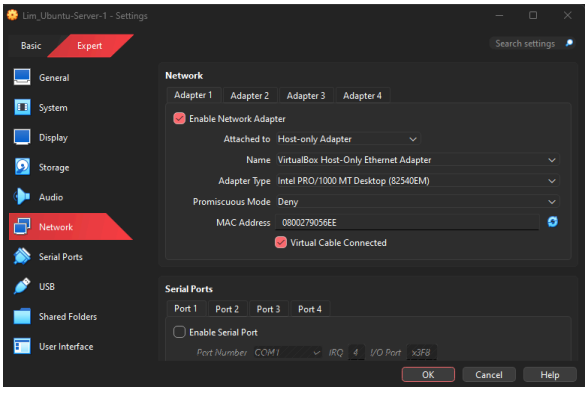
Task 3: Verify network settings on Server 1, Server 2, and Local Machine.

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check



network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.





In this part I refreshed the MAC Address of each VM for them to have different IP addresses.

1.1 WorkstationIP address: 192.168.56.6

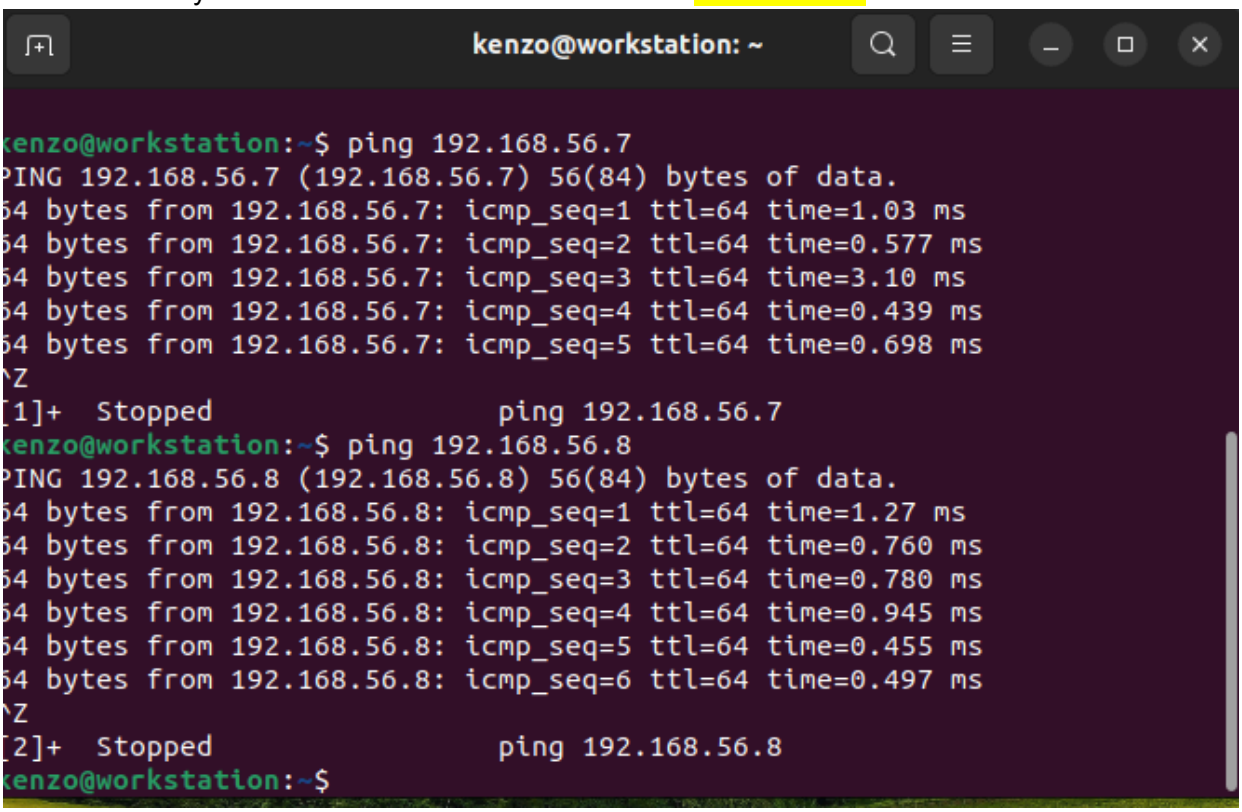
1.2 Server 1 IP address: 192.168.56.7

1.3 Server 2 IP address: 192.168.56.8

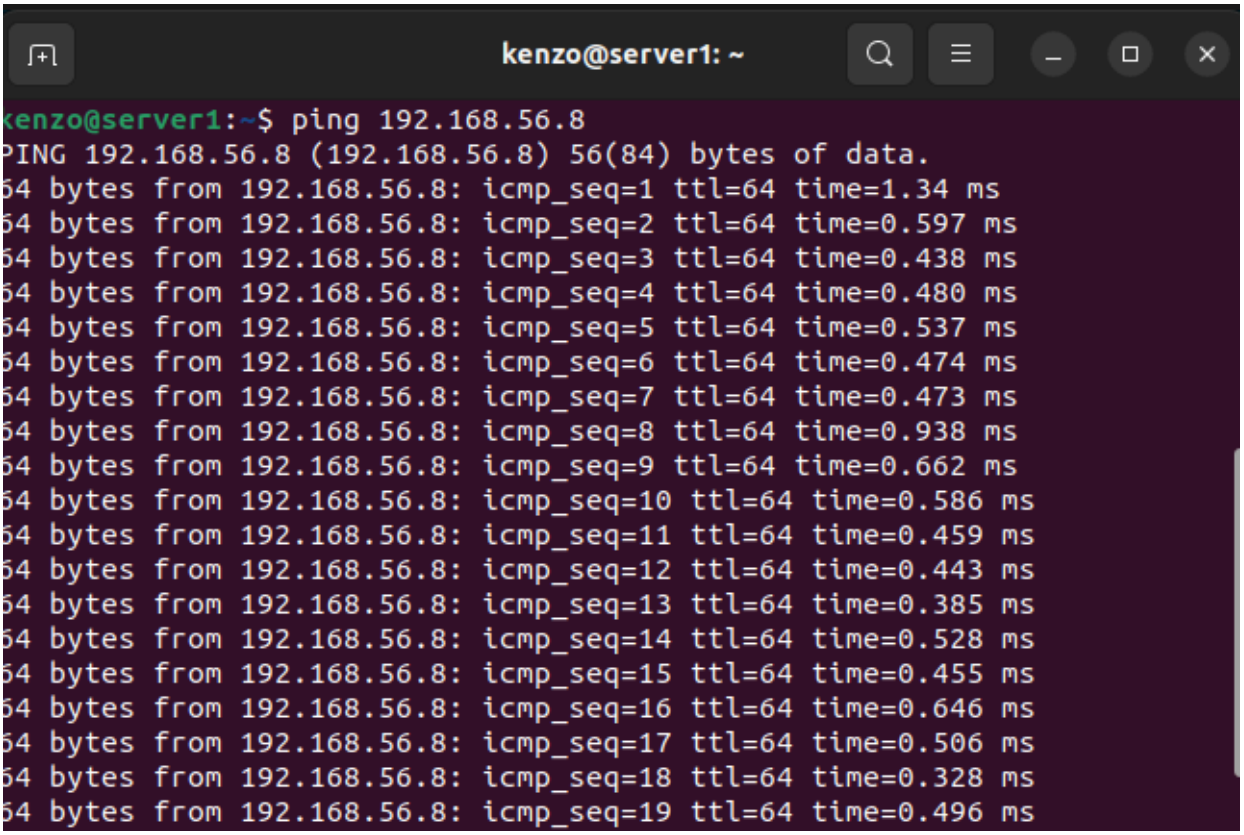
2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: **/ Successful** **X** Not Successful

2.2 Connectivity test for Local Machine 1 to Server 2: **/ Successful** **X** Not Successful



2.3 Connectivity test for Server 1 to Server 2: **/ Successful** **X** Not Successful



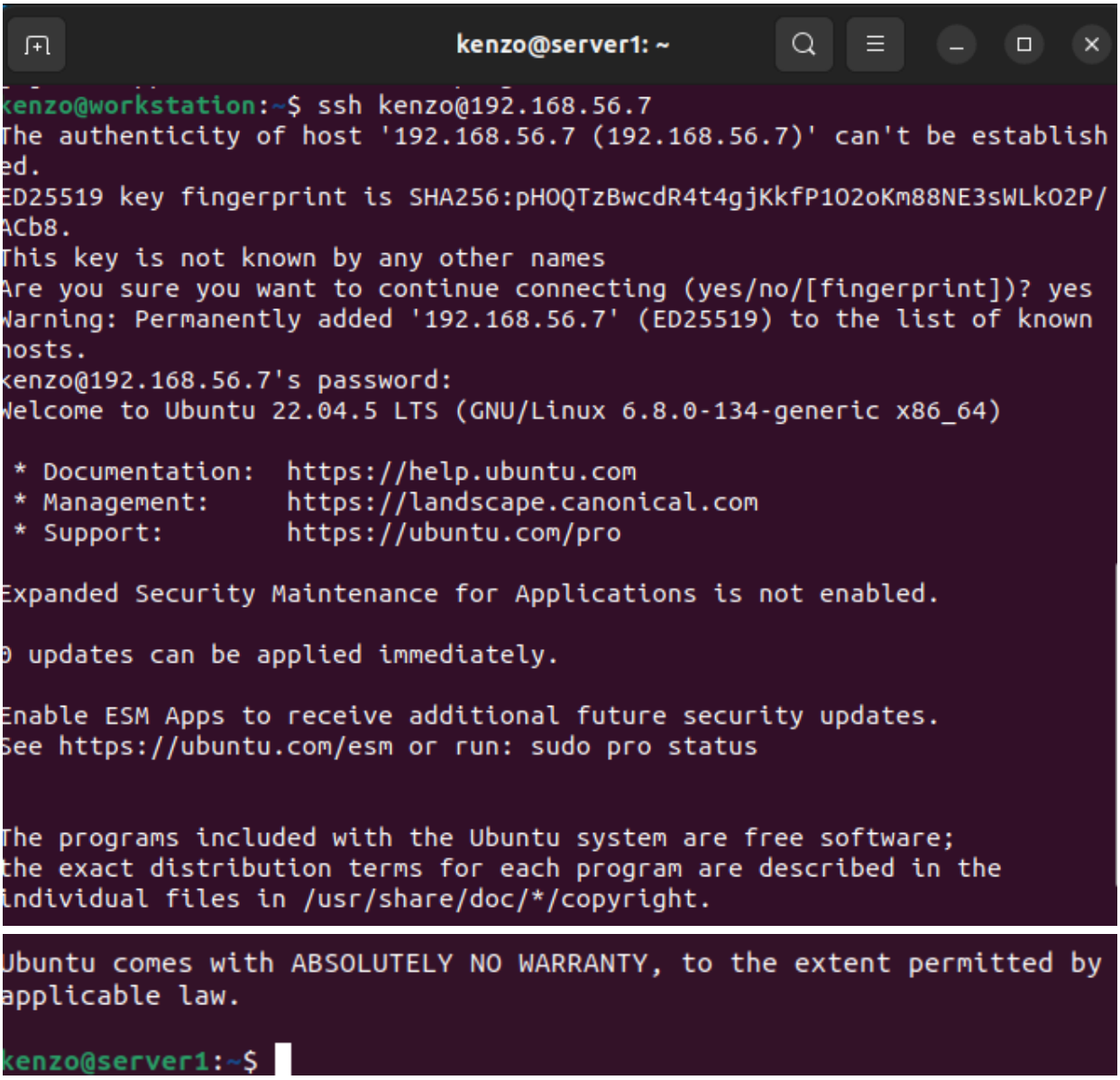
Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:
1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120

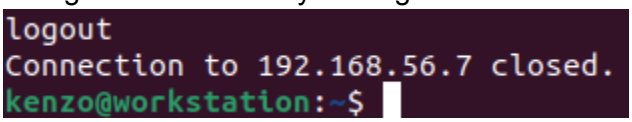
1.2 Enter the password for server 1 when prompted

1.3 Verify that you are in server 1. The user should be in this format user@server1.

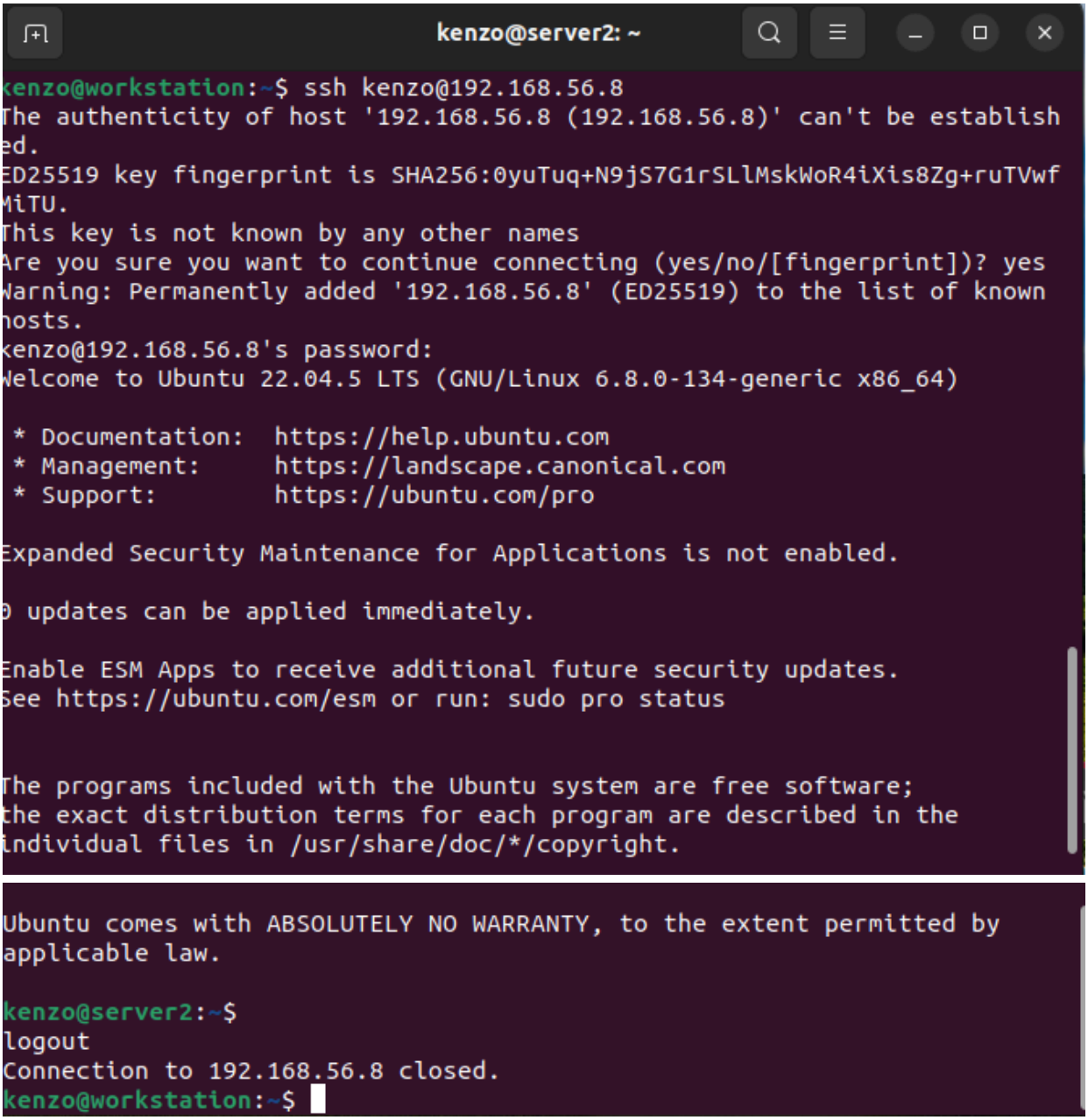
For example, jvtaylar@server1



2. Logout of Server 1 by issuing the command control + D.



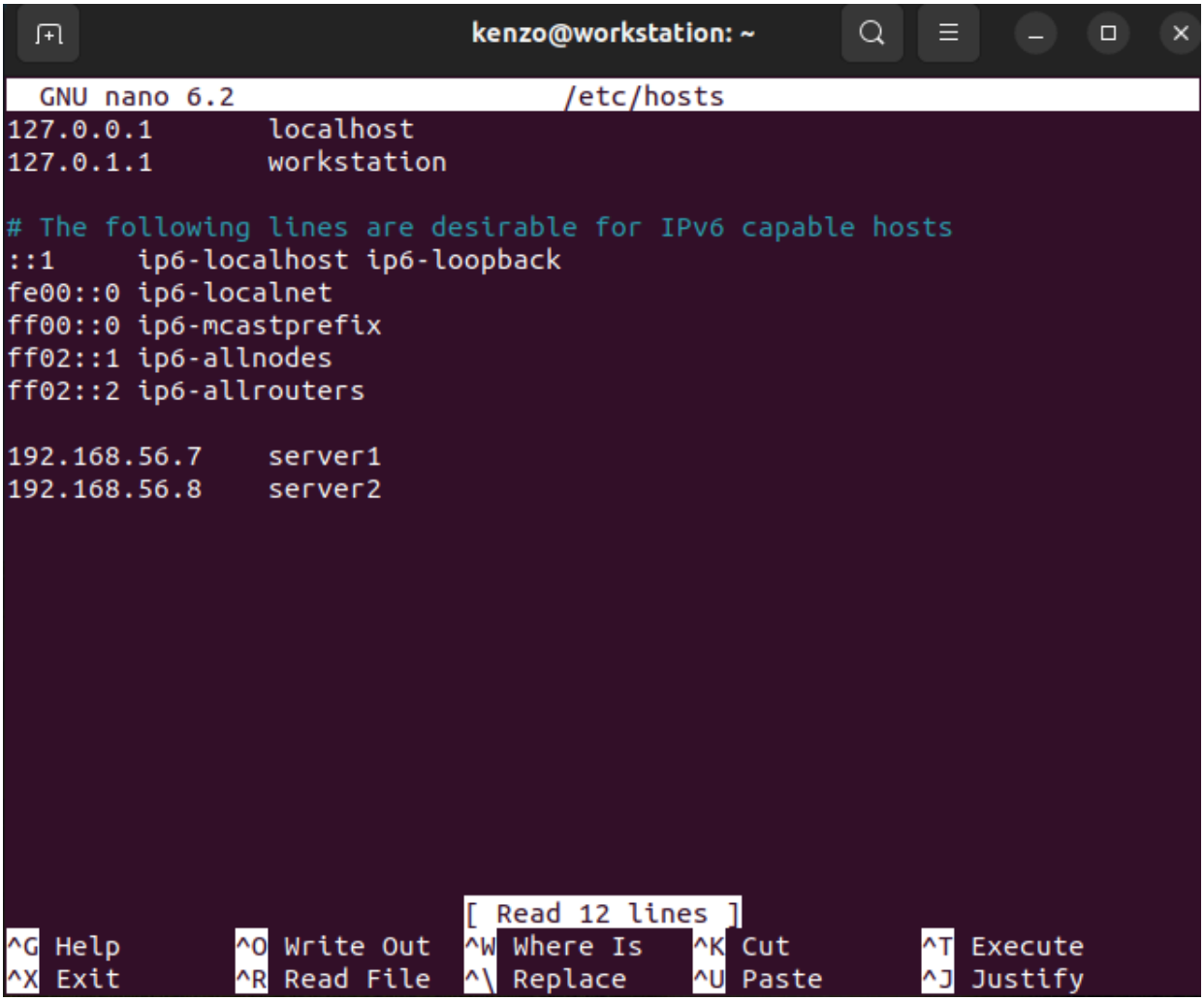
3. Do the same for Server 2.



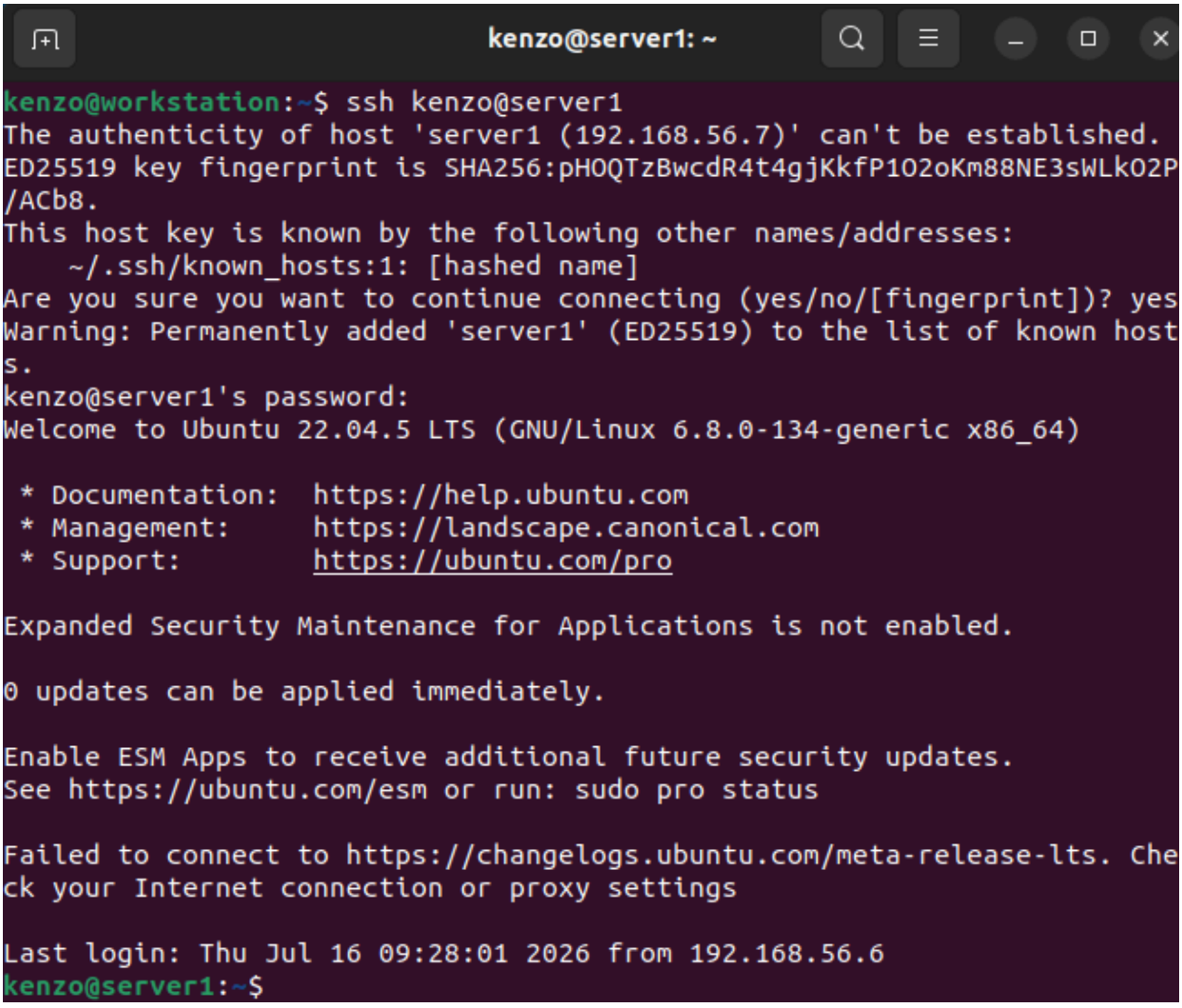
4. Edit the hosts of the Local Machine by issuing the command sudo nano/etc/hosts. Below all texts type the following:
4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)

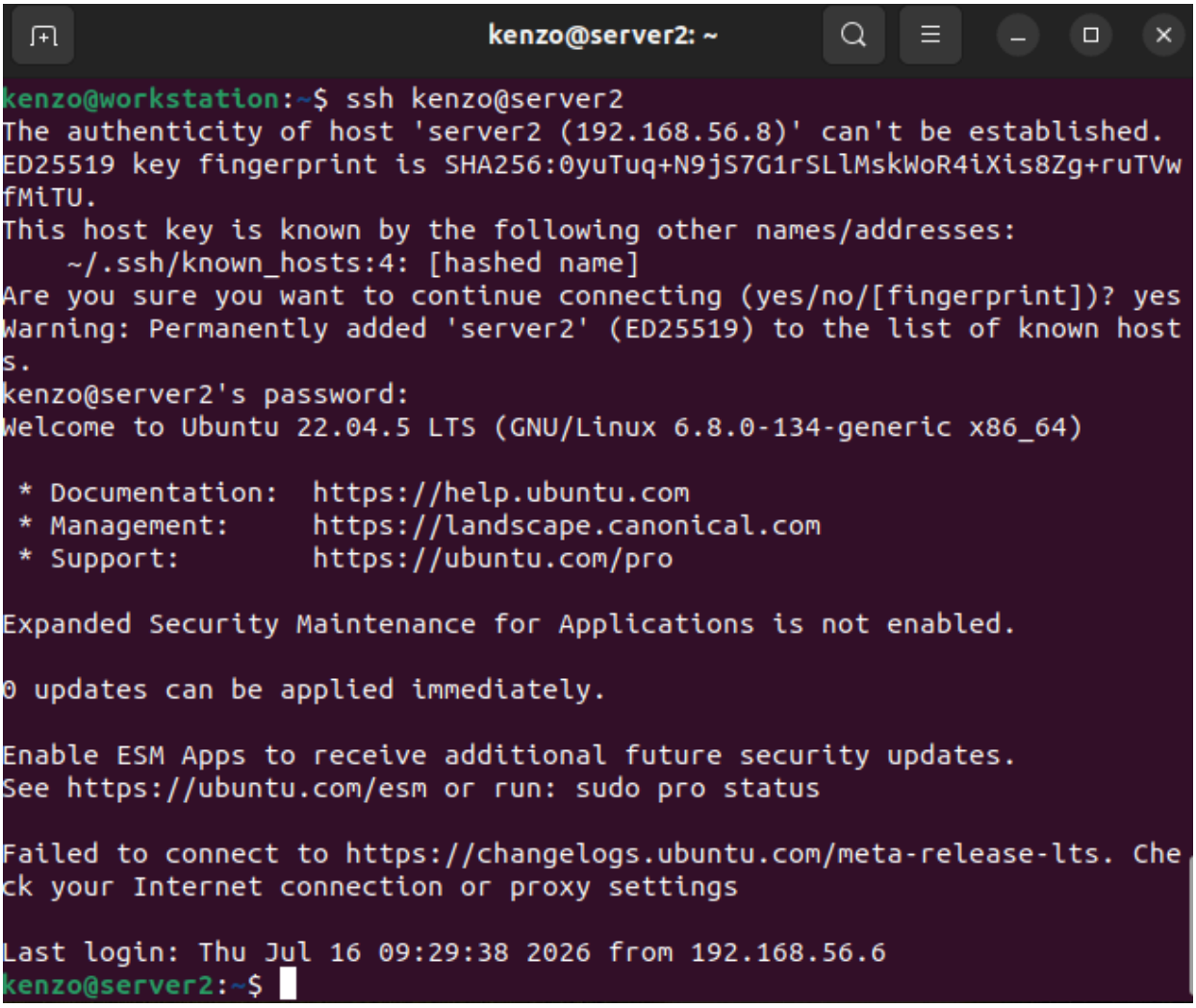
4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

4.3 Save the file and exit.



5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password
when prompted. Verify that you have entered Server 1. Do the same for Server 2


kenzo@server1: ~
kenzo@workstation:~$ ssh kenzo@server1
The authenticity of host 'server1 (192.168.56.7)' can't be established.
ED25519 key fingerprint is SHA256:pHOQTzBwcdR4t4gjKkfP1O2oKm88NE3sWLkO2P
/ACb8.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known host
s.
kenzo@server1's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Che
ck your Internet connection or proxy settings
Last login: Thu Jul 16 09:28:01 2026 from 192.168.56.6
kenzo@server1:~$



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.
Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?

In this case, we configured the /etc/hosts and added the IP and hostnames of servers 1 and 2, which I suppose that the system automatically uses the saved hosts inside it and for example if i run the ssh user@hostname it fetches it inside the hosts directory which we modified.

2. How secured is SSH?

SSH is highly secure, but its actual security depends on how you configure it. It has features of strong encryption, cryptographic authentication, and data integrity. For strong encryption, all data moving between the client and the server are scrambled using symmetric encryption like AES, cryptographic authentication as it relies on asymmetric public-private key pairs to verify the identity of the client and the host. Lastly the data integrity, it utilizes hashing algorithms like SHA to verify that packets have not been altered.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Concept explanation**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ChatGPT to ask whether I can ping cloned VMs and also asked about SSH features.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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